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RESEARCH TOPIC

Recent developments in generative AI for protein structure prediction and drug discovery

EXECUTIVE SUMMARY

Topic Framing

Recent advancements in generative AI have significantly impacted protein structure prediction and drug discovery. These developments leverage deep learning and machine learning to enhance the accuracy and efficiency of predicting protein structures and identifying potential drug candidates. The retrieved articles provide insights into these advancements, highlighting both the progress made and the challenges that remain.

Most Relevant Papers and Main Contributions

1. **"Deep learning revolutionizes protein research: Advances in structure prediction, functional annotation, and engineered design"**: This paper underscores the transformative impact of deep learning models like AlphaFold2 on protein structure prediction. These models have achieved near-experimental accuracy for single-domain, globular proteins with sufficient evolutionary data, creating a vast repository of high-confidence predicted structures. This advancement addresses a core challenge in structural biology and facilitates further research in functional annotation and protein design.

2. **"More protein-ligand data are needed for AlphaFold-like models to enable drug discovery"**: This article highlights the potential of protein structure prediction models (PSPMs) in structure-based drug design (SBDD). However, it emphasizes the need for more comprehensive datasets and blind challenges to assess and improve the models' applicability to real-world drug discovery tasks.

3. **"Computational and AI-Driven Ecosystem for Structure-Based Covalent Drug Discovery"**: The paper discusses the resurgence of covalent drug discovery, driven by AI and computational methods. It emphasizes the importance of creating a synergistic ecosystem that integrates data-driven approaches to enhance the development of covalent drugs.

4. **"Pocket-Based Generative Diffusion Model Accelerates Potent Influenza A Hemagglutinin Inhibitor Discovery"**: This study introduces a dual conditional diffusion model (DCDM) that refines 3D

target-based molecular generation by leveraging ligand-protein interaction features. The model improves predicted binding affinity while maintaining structural rationality and diversity, showcasing its potential in optimizing drug candidates.

Recurring Trends and Methodological Patterns

- **Integration of AI and Machine Learning**: Across the articles, there is a clear trend of integrating AI and machine learning with traditional modeling approaches to enhance the prediction of protein structures and drug interactions. This integration is seen in the use of deep learning models for protein structure prediction and machine learning-enhanced QSAR models for drug discovery.
- **Need for Comprehensive Datasets**: Several papers emphasize the necessity of expanding existing datasets to improve the applicability and accuracy of AI models in drug discovery. This is particularly crucial for tasks that require real-world applicability, such as SBDD.
- **Focus on Structure-Based Approaches**: There is a strong focus on structure-based approaches in drug discovery, leveraging the detailed structural information provided by advanced AI models to design and optimize therapeutic candidates.

Limitations and Evidence Gaps

- **Data Limitations**: A significant limitation highlighted is the lack of comprehensive protein-ligand interaction data, which is crucial for improving the applicability of AI models in drug discovery.
- **Model Applicability**: While AI models have shown promise in controlled settings, their performance in real-world drug discovery tasks remains uncertain. The need for blind challenges and benchmarking is emphasized to better understand model limitations.
- **Complexity of Biological Systems**: The complexity and variability of biological systems pose challenges in accurately predicting drug interactions and efficacy, indicating a need for more sophisticated models and diverse datasets.

Conclusion

The retrieved articles collectively illustrate the significant advancements in generative AI for protein structure prediction and drug discovery. While these technologies have achieved remarkable progress, particularly in structure prediction, challenges remain in terms of data availability and model applicability to real-world tasks. Addressing these limitations through expanded datasets and rigorous benchmarking will be crucial for realizing the full potential of AI in drug discovery.

1

Deep learning revolutionizes protein research: Advances in structure prediction, functional annotation, and engineered design.

PubMed | 2026-03-14 | DOI: 10.1016/j.jbiotec.2026.03.012

Abstract. Recent advances in deep learning have fundamentally transformed protein research by creating a synergistic cycle that connects structure prediction, functional annotation, and rational design. This review presents an integrative framework to demonstrate how breakthroughs in one domain catalytically enable progress in the next. First, for a broad range of single-domain, globular proteins-particularly those with sufficient evolutionary information-end-to-end deep learning models exemplified by AlphaFold2 have achieved near-experimental accuracy, effectively addressing a core challenge in structural biology and generating an unprecedented repository of high-confidence predicted structures. This vast structural substrate then serves as the essential foundation for the "Understand" phase, where multimodal models increasingly integrate 3D coordinates with sequence and interaction data to achieve precise, mechanism-aware functional predictions that transcend homology-based inference. These deep functional insights, in turn, provide the critical biochemical constraints for the final "Create" phase. Here, generative AI and inverse folding models enable the de novo design of novel proteins-f

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2

Integration of Machine Learning With PBPK and QSAR Modeling Approaches to Facilitate Drug Discovery and Development.

PubMed | 2026-03-01 | DOI: 10.1002/psp4.70228

Abstract. This review examines the application of machine learning (ML) in physiologically based pharmacokinetic (PBPK) modeling through improved prediction of input parameters, particularly via quantitative structure-activity relationship (QSAR) models, for absorption, distribution, metabolism, and excretion (ADME) properties across drug development phases. Traditional PBPK models, while mechanistically sound, face limitations in applicability domain when compound-specific physicochemical and biochemical parameters (e.g., partition coefficients, metabolic clearance rates, plasma protein binding) cannot be reliably determined or estimated for novel chemicals. ML-enhanced QSAR models address these input parameter limitations by predicting required parameters directly from chemical structure, thereby extending PBPK applicability to diverse chemical space without altering the mechanistic model framework itself. Recent advances in ML-powered QSAR methodology generate higher-quality predictions of partition coefficients, clearance rates, and binding parameters. When these ML-predicted parameters are integrated as inputs into conventional PBPK frameworks, the resulting simulations often demonstrat

Source URL. https://pubmed.ncbi.nlm.nih.gov/41878892

3 More protein-ligand data are needed for AlphaFold-like models to enable drug discovery.

PubMed 2026-03-31 DOI: 10.1016/j.sbi.2026.103257

Abstract. Structure-based drug design (SBDD) is an evolving paradigm that leverages protein structural information to improve small molecule therapeutic design. Building on more than 50 years of data curation from the Protein Data Bank, the recent emergence of protein structure prediction models (PSPMs) promises to enable new computationally driven approaches for therapeutic discovery. However, it is critical to assess the limitations of these models using blind challenges and to expand existing datasets to better reflect real-world drug design tasks. Here, we discuss recent efforts to benchmark existing PSPMs and identify their limitations. We offer a hierarchical framework for parsing which tasks the current models perform well, and which tasks remain challenging or unexplored. Finally, we emphasize the need for systematic dataset generation to support the development of frontier models and highlight recent efforts to generate experimental and physics-based datasets for challenging tasks in drug discovery.

Source URL. https://pubmed.ncbi.nlm.nih.gov/41924828

4 Recent advances in computational antimicrobial peptide discovery through big data, modeling, and artificial intelligence and their interplay in ushering the next golden era of drug development.

EuropePMC 2026-03-17 DOI: 10.3389/fbinf.2026.1749404

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41923798

5 Computational and AI-Driven Ecosystem for Structure-Based Covalent Drug Discovery.

PubMed 2026-02-27 DOI: 10.1021/acs.accounts.5c00905

Abstract. ConspectusThe field of covalent drug discovery has witnessed a remarkable resurgence in recent years, a trend underscored by the approval of more than 125 covalent drugs by the US FDA as of 2025, which demonstrates their immense therapeutic potential. Driven by ever-increasing computational power and vast amounts of data, deep learning (DL) is profoundly transforming numerous fields, from natural language processing to drug discovery. In the development of covalent drugs, in particular, advanced computational methods centered on data-driven approaches and artificial intelligence (AI) exhibit immense potential. The realization of this potential depends on the construction of a synergistic ecosystem. Here, we define this "ecosystem" as an integrated set of components—including (i) curated covalent-relevant databases, (ii) AI/physics-based predictive and scoring models, (iii) interoperable computational workflows spanning site identification, docking/virtual screening, and lead optimization, and (iv) closed-loop feedback that systematically incorporates experimental outcomes to update data resources and refine/validate models. This begins with the systematic collection of past experim

Source URL. https://pubmed.ncbi.nlm.nih.gov/41758998

6 ERK1/2-targeted Cancer therapies: Recent advances, potential drug resistance, and applicability analysis of emerging technologies.

PubMed | 2026-03-19 | DOI: 10.1016/j.bioorg.2026.109775

Abstract. As the terminal effector of the RAS-RAF-MEK-ERK signaling cascade, extracellular signal-regulated kinase 1/2 (ERK1/2) plays a central role in transmitting oncogenic signals, and its dysregulated activation directly drives tumor initiation, disease progression, and poor clinical outcomes. Although inhibitors targeting upstream nodes, such as RAF and MEK have demonstrated clinical benefit, the frequent emergence of acquired resistance remains a major barrier, ultimately limiting their long-term therapeutic effectiveness. Consequently, direct inhibition of ERK1/2 has gained increasing attention as a promising strategy, offering not only potent suppression of MAPK pathway activity but also the potential to overcome resistance driven by upstream genetic alterations. Despite its therapeutic promise, the clinical development of ERK1/2 inhibitors is confronted with several significant challenges. Firstly, although first-generation ATP-competitive inhibitors such as Ulixertinib have shown antitumor activity in early-phase trials, their efficacy varies widely across patient populations and is often accompanied by mechanism-based toxicities (including rash and diarrhea), highlighting a substa

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7 AI-driven drug-target interaction prediction: current progress, challenges, and future roadmap for precision medicine.

PubMed | 2026-03-31 | DOI: 10.1007/s10822-026-00798-2

Abstract. Drug-target interactions (DTIs) are fundamental to drug discovery, development, and repositioning. However, experimental methods for DTI identification are often constrained by high costs, time demands, and scalability issues, prompting a shift toward computational approaches. This review systematically explores recent advancements in computational DTI prediction, encompassing ligand-based, target-based, network-based, machine learning (ML), deep learning (DL), and hybrid multi-omics models. Ligand-based techniques, such as QSAR and pharmacophore modeling, offer structure-activity insights but require known ligands. Target-based methods rely on molecular docking and binding site prediction, yet often suffer from incomplete or unknown protein structures. Network-based strategies utilize bipartite and heterogeneous graphs integrated with protein-protein interaction (PPI) networks to infer novel DTIs. ML and DL methods especially graph neural networks and Transformer-based models have significantly improved prediction accuracy by leveraging chemical, biological, and omics features. Notably, hybrid models that integrate genomics, transcriptomics, proteomics, and interactomics data offe

Source URL. https://pubmed.ncbi.nlm.nih.gov/41910880

8 Generative AI in drug repurposing and biomarker discovery: a multimodal approach.

EuropePMC | 2026-03-20 | DOI: 10.3389/fbinf.2026.1755412

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41938337

9 **Protein structure prediction powered by artificial intelligence: from biochemical foundations to practical applications.**

EuropePMC | 2026-03-09 | DOI: 10.3389/fmolb.2026.1767821

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41878008

10 **Next-Generation Artificial Intelligence for ADME Prediction in Drug Discovery: From Small Molecules to Biologics.**

EuropePMC | 2026-02-15 | DOI: 10.33160/yam.2026.02.001

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41709935

11 **Pocket-Based Generative Diffusion Model Accelerates Potent Influenza A Hemagglutinin Inhibitor Discovery.**

PubMed | 2026-02-13 | DOI: 10.1021/acs.jmedchem.5c03787

Abstract. The deep generative model has recently advanced 3D chemical space exploration but overlooked the balance between target affinity and structural rationality, limiting their effectiveness in drug discovery. Herein, we established a novel dual conditional diffusion model (DCDM) that leveraged ligand-protein interaction features to refine 3D target-based molecular generation. DCDM exhibited superiority in enhancing predicted binding affinity while maintaining high structural rationality and diversity. Subsequently, we applied DCDM to optimize penindolone (PND), a marine-derived lead compound from our laboratory, targeting influenza A hemagglutinin (HA). Efficiently, a promising candidate (compound

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12 **Foundation and Multimodal Models for Drug Discovery in Molecular Informatics: Principles, Evaluation, and Practical Guidance.**

EuropePMC | 2026-03-01 | DOI: 10.1002/minf.70027

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41877557

13 **Peptide-based drug design using generative AI.**

EuropePMC | 2026-01-13 | DOI: 10.1039/d5cc04998a

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41376388

14 Artificial Intelligence for Natural Products Drug Discovery in Neurodegenerative Therapies: A Review.

EuropePMC | 2026-01-12 | DOI: 10.3390/biom16010129

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41594669

15 Structure-informed machine learning for drug discovery: a task-centric perspective.

EuropePMC | 2026-01-01 | DOI: 10.1093/bib/bbag081

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41729820

16 A review of recent advances in generative artificial intelligence models for biomolecular sciences.

EuropePMC | 2025-12-11 | DOI: 10.1016/j.apsb.2025.12.012

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41909740

17 Not Only are Protectionist Measures Harmful for Economic Development but the Recent Changes in the Global Trading System have Rendered them Irrelevant

CrossRef | 2014-02-05 | DOI: 10.2139/ssrn.2389355

Abstract. No abstract available.

Source URL. https://doi.org/10.2139/ssrn.2389355

18

Chemico-biological evaluation of carpachromene against key antimicrobial protein targets: an integrated in-silico, in-vitro approach for mechanistic insights

OpenAlex 2026-04-17 DOI: https://doi.org/10.3389/fphar.2026.1785267

Abstract. Ethnopharmacological Relevance Verbascum thapsus L. is a prized medicinal plant from the Kashmir Himalaya traditionally utilized to treat many ailments, yet its active metabolites against antimicrobial mechanisms remain unclear. Aim of the Study This work envisages an integrated in-silico and in-vitro approach to mechanistically investigate broad-spectrum antimicrobial activity of carpachromene, a supradecorated phytochemical from V . thapsus . Materials and Methods Carpachromene was isolated through cold extraction from V . thapsus , followed by silica gel column chromatography with an optimized polarity solvent system, yielding a whitish amorphous solid confirmed by XRD, FTIR, 1 H NMR and 13 C NMR spectroscopy. In-silico analyses encompassed molecular docking of carpachromene against key microbial drug targets like sterol 14-α demethylase (CYP51), Dihydropteroate synthase (DHPS), GyrB ATPase domain, and Penicillin-Binding Protein 1 (PBP-1) followed by 100 ns molecular dynamics simulations assessing RMSD, RMSF, dynamic cross-correlation matrix (DCCM), principal component analysis (PCA), radius of gyration (Rg), and solvent accessible surface area (SASA). In-vitro antimicrobial act

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19

Advances in multi-omics research on neuroblastoma

OpenAlex 2026-04-16 DOI: https://doi.org/10.3389/fped.2026.1739376

Abstract. Neuroblastoma is the most common extracranial solid tumor in children, presenting significant challenges in diagnosis and treatment due to its highly heterogeneous clinical manifestations and complex genetic background. In recent years, advances in transcriptomics have played a pivotal role in this field, not only aiding in the identification of molecular subtypes of tumors but also revealing potential mechanisms of drug resistance. Through comprehensive gene expression profiling and single-cell sequencing technology, researchers have deeply analyzed key interaction nodes within the metabolic-immune microenvironment, providing a theoretical basis for developing targeted therapeutic strategies. Concurrently, radiomics, leveraging imaging techniques such as MRI, PET-CT, and CT, quantitatively assesses the morphological and metabolic characteristics of tumors. This enables non-invasive prediction of MYCN amplification status, evaluation of bone marrow metastasis risk, and prognostic stratification, thereby supporting dynamic disease monitoring. In pathology, artificial intelligence technology is widely applied in the analysis of digital pathology images. It effectively identifies cell

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20 Discovery of co-stimulatory anti-CD28 VHHs for developing cancer immune therapeutic anti-tumor/CD3/CD28 trispecific T cell engager

OpenAlex 2026-04-17 DOI: https://doi.org/10.3389/fimmu.2026.1812063

Abstract. Introduction CD28 is a critical co-stimulatory receptor expressed on T cells. Alongside the T-cell receptor (TCR), it provides a dual-signal requirement that is necessary for the complete activation of T cells. Developing co-stimulatory agonists targeting CD28 therefore represents a promising strategy to enhance the efficacy of T cell based anti-tumor immunotherapy. Results In this study, we isolated 34 VHHs (VHHs) from a phage-display immune library generated using PBMCs of a Bactrian camel immunized with recombinant human CD28 protein. These VHHs bound to cell surface CD28 with EC 50 values ranging from 8.5 nM to 130 nM and exhibited varied co-stimulatory activity. When incorporated as building blocks into a trispecific T cell engager (Tri-TCE) targeting DLL3, CD3, and CD28, the anti- CD28 VHHs conferred superior tumor specific cytotoxicity compared to a bispecific T cell engagers (Bi-TCEs) targeting only DLL3 and CD3. Among all candidates, the Tri-TCE containing VHH38 demonstrated the most potent antitumor efficacy. Structural mapping and alanine-scanning mutagenesis revealed that VHH38 engages the CD28 epitope, involving residues Arg34, Tyr51, and Tyr61, a region similar to tha

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21 A Validated Low-to-Intermediate Mass Planetary Interior Structure Model and New Mass-Radius Relations

arXiv 2026-04-16

Abstract. The increasing precision of planetary mass and radius observations is bringing major questions about the structure and formation of planets--such as the nature of the radius valley and origin of super-Mercuries--within reach, demanding the development of interior structure models with more physics to more accurately determine planetary radii for a given composition. Here, we present a new model that includes state-of-the-art equations of state following the latest experimental and computational results, a physically-motivated mineralogy allowing multiple species to coexist within planetary layers, a non-adiabatic temperature profile, melting, and other features. This model replicates Earth's radius and moment of inertia coefficient to within 0.2%, Mars and the Moon's to within 0.5%, and Mercury, Venus, and Europa's to within 1% or 3$σ$. We use this model to calculate mass-radius relationships for H/He-enveloped, water-rich, Earth-like, and iron-rich bodies with masses between $0.01-100\, M_\oplus$. We calculate mass-radius tables and fit piece-wise power-laws to them for ${<}8M_\oplus$ planets, finding that the exponent in $M=aR^b$ increases with mass and core mass fractio

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22 Knowledge discovery and computerized reasoning to assist tourist destination marketing

CrossRef | This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.15556/ijsim.01.02.004

Abstract. No abstract available.

Source URL. https://doi.org/10.15556/ijsim.01.02.004

23 Fish assemblage composition and structure in a pampean stream with contrasting land uses.

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DOI: 10.22179/revmacn.20.481

Abstract. No abstract available.

Source URL. https://doi.org/10.22179/revmacn.20.481

24 Education and Social Structure in a Restructuring Economy: Vietnam as Case Study

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DOI: 10.20431/2349-0381.0303001

Abstract. No abstract available.

Source URL. https://doi.org/10.20431/2349-0381.0303001

25 The crossed-product structure of C*-algebras arising from topological dynamical systems

CrossRef | This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.7900/jot.2011may31.1924

Abstract. No abstract available.

Source URL. https://doi.org/10.7900/jot.2011may31.1924

26

Structure determination of five trace chemical constituents from roots of Linum usitatissimum

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DOI: 10.4268/cjcmm20110215

Abstract. No abstract available.

Source URL. https://doi.org/10.4268/cjcmm20110215

27

Determination and comparison of plasma protein binding rate of alkaloids from seed of Strychnou nux-vomica

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This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.4268/cjcmm20110221

Abstract. No abstract available.

Source URL. https://doi.org/10.4268/cjcmm20110221

28

USING OF LAB ANIMALS AS A MODEL TO STUDY THE DRUG TOXICITY IN COVID-19

CrossRef

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DOI: 10.23975/bjvetr.2031.175853

Abstract. No abstract available.

Source URL. https://doi.org/10.23975/bjvetr.2031.175853

29

The Development of Loyalty with Online Services Customers

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.15556/ijsim.04.03.001

Abstract. No abstract available.

Source URL. https://doi.org/10.15556/ijsim.04.03.001

30 Corporate Social Responsibility: A marketing tool and/or a factor for the promotion of sustainable development for companies? An empirical examination of listed companies in the Athens Stock Exchange

CrossRef | This is an accepted article with a DOI pre-assigned that is not yet published. | DOI: 10.15556/ijsim.02.01.002

Abstract. No abstract available.

Source URL. https://doi.org/10.15556/ijsim.02.01.002

31 Repurposing AI for protein interactions and dynamics: opportunities, limitations, and lessons.

EuropePMC | 2026-03-11 | DOI: 10.3389/fbinf.2026.1749317

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41889389

32 Artificial Intelligence-Driven Discovery and Optimization of Antimicrobial Peptides Targeting ESKAPE Pathogens and Multidrug-Resistant Fungi.

EuropePMC | 2026-03-06 | DOI: 10.3390/microorganisms14030591

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41900351

33 AI accelerate the identification of druggable targets by 3D structures of proteins and compounds.

EuropePMC | 2026-02-14 | DOI: 10.1038/s41698-026-01310-7

Abstract. No abstract available.

Source URL. https://europepmc.org/article/MED/41691035

34 Combating Antiviral Drug Resistance: A Multipronged Strategy.

PubMed 2026-02-06 DOI: 10.1021/acs.accounts.5c00724

Abstract. ConspectusViral proteases are essential enzymes required for viral replication and assembly, making them prime antiviral drug targets. However, under the selective pressure of protease inhibitors, viruses can acquire mutations that reduce drug binding efficacy, posing significant challenges in both chronic infections (e.g., HIV, HCV) and acute infections like COVID-19, where mutations in the SARS-CoV-2 main protease (M

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35 Inflammasomes meet organoids and artificial intelligence: unraveling the complexity of gynecological inflammation

OpenAlex 2026-04-17 DOI: https://doi.org/10.3389/fimmu.2026.1753651

Abstract. Gynecological diseases represent a persistent global health burden. According to a WHO report, the global incidence of gynecological diseases exceeds 65%. Furthermore, over 90% of women suffer from gynecological issues to varying degrees. They account for about 4.5% of the global disease burden. This share is higher than that of other major global health problems. For example, malaria accounts for about 1.04%. Tuberculosis accounts for about 1.9%. Ischemic heart disease accounts for about 2.2%. Mounting evidence suggests that inflammasomes act as master regulators linking chronic inflammation with the onset and progression of gynecological disorders including endometriosis, ovarian cancer, and polycystic ovary syndrome. Yet, traditional two-dimensional cultures and animal models fail to reproduce the intricate immune–endocrine microenvironment of the human reproductive system, impeding translational progress. This review provides an integrated perspective that unites inflammasome biology, organoid technology, and artificial intelligence (AI) into a new research paradigm for precision immunology. We comprehensively summarize the molecular mechanisms of inflammasome activation—partic

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36

Molecular Modeling of the Pathogenetic Mechanisms of Neuropsychiatric Disorders

OpenAlex 2026-04-16 DOI: https://doi.org/10.3390/ijms27083563

Abstract. Neuropsychiatric diseases are characterized by complex molecular underpinnings that remain challenging to fully elucidate. Molecular dynamics (MD) simulations have emerged as a powerful computational tool, providing a crucial bridge between static genetic data and the dynamic functional consequences of molecular alterations. This review offers a comprehensive overview of the application of MD simulations in studying the molecular basis of neuropsychiatric disorders. We highlight key applications, including the assessment of mutation pathogenicity in disease-associated proteins, the influence of post-translational modifications on protein function, folding, misfolding, and aggregation, and the characterization of psychopharmacological drug–target interactions at atomic resolution. Through relevant examples from research on psychiatric and neurodegenerative diseases, we illustrate how these computational methods are implemented to gain mechanistic insights. Importantly, this review traces the historical development of MD simulations in biological applications, critically examines the method's limitations, and outlines future perspectives for simulating long-timescale physiological pr

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37

The neuro-immune axis in preeclampsia: from the maternal-fetal interface to systemic dysregulation

OpenAlex 2026-04-17 DOI: https://doi.org/10.3389/fimmu.2026.1761197

Abstract. Preeclampsia (PE) is a complex hypertensive disorder of pregnancy characterized by new-onset maternal hypertension and multi-organ dysfunction. Although placental maladaptation and immune activation are well-established features of PE, growing evidence indicates that dysregulated neuro–immune–vascular integration critically contributes to disease initiation, progression, and long-term sequelae. Normal pregnancy requires coordinated immune and neural adaptations, particularly at the maternal–fetal interface, to support successful placentation. While placental pathology, angiogenic imbalance, and immune activation establish the systemic environment of PE, some neurological phenotypes (such as eclampsia and acute cerebral autoregulatory failure) are difficult to explain without involvement of central autonomic and sensory integration circuits that mediate the translation of peripheral inflammatory and vasoactive signals into neurovascular responses. Dysfunction of cerebral autoregulation has been proposed as a key mechanism underlying acute neurological complications, independent of classic placental factors. In PE, this finely tuned communication becomes spatially and functionally di

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38

Energy metabolism dysregulation in idiopathic inflammatory myopathies: mechanisms and therapeutic implications

OpenAlex | 2026-04-17 | DOI: https://doi.org/10.3389/fimmu.2026.1781434

Abstract. Idiopathic inflammatory myopathies (IIMs) are being increasingly recognized as disorders driven by profound disturbances in cellular energy metabolism rather than inflammation alone. Recent studies have highlighted mitochondrial dysfunction, oxidative stress, and metabolic reprogramming across glucose, lipid, and amino acid pathways as central mechanisms linking energy metabolism dysregulation to sustained muscle injury. Defective mitophagy, mitochondrial DNA (mtDNA) depletion, and excessive reactive oxygen species (ROS) production create a self-amplifying loop with interferon-driven inflammation, whereas abnormal glycolysis, impaired fatty acid oxidation, and dysregulated tryptophan–kynurenine metabolism further shape the immunometabolic landscape of IIMs. These metabolic shifts not only contribute to muscle weakness and tissue degeneration but are also correlated with disease severity, autoantibody profiles, and treatment resistance. Emerging therapeutic strategies, including antioxidant approaches, mitochondrion-targeted agents, metabolic modulators, and exercise-based interventions, underscore the translational potential of targeting energy homeostasis. This review synthesizes

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39

LeapAlign: Post-Training Flow Matching Models at Any Generation Step by Building Two-Step Trajectories

arXiv | 2026-04-16

Abstract. This paper focuses on the alignment of flow matching models with human preferences. A promising way is fine-tuning by directly backpropagating reward gradients through the differentiable generation process of flow matching. However, backpropagating through long trajectories results in prohibitive memory costs and gradient explosion. Therefore, direct-gradient methods struggle to update early generation steps, which are crucial for determining the global structure of the final image. To address this issue, we introduce LeapAlign, a fine-tuning method that reduces computational cost and enables direct gradient propagation from reward to early generation steps. Specifically, we shorten the long trajectory into only two steps by designing two consecutive leaps, each skipping multiple ODE sampling steps and predicting future latents in a single step. By randomizing the start and end timesteps of the leaps, LeapAlign leads to efficient and stable model updates at any generation step. To better use such shortened trajectories, we assign higher training weights to those that are more consistent with the long generation path. To further enhance gradient stability, we reduce the weights of g

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40

Impact of the halogen PB radii in the estimation of protein-ligand binding energies using MM-PBSA calculations.

PubMed 2026-04-16 DOI: 10.1039/d5cp03537f

Abstract. Halogenation is a widely used strategy in drug design, not only to improve ADME properties but also because halogens can form halogen bonds (XBs) with biological targets. To predict protein-ligand binding free energies (Δ

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